KLE Technological University
Hubballi

A Project Report on

"BloodBond"

A Project Report Submitted in Partial Fulfillment of the Requirement for the Course of

Minor Project-1 (23ECSW303)

in

6th Semester of Computer Science and Engineering

by

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Belagavi – 590 008.

June 2024

DECLARATION

We hereby declare that the matter embodied in this report entitled "**BloodBond**" submitted to KLE Technological University for the course completion of Minor Project-1 (23ECSW303) in the 6^{th} Semester of Computer Science and Engineering is the result of the work done by us in the Department of Computer Science and Engineering, KLE Technological University's Dr. M. S. Sheshgiri College of Engineering, Belagavi under the guidance of Prof.Savita Bagewadi, Department of Computer Science and Engineering. We further declare that to the best of our knowledge and belief, the work reported here in doesn't form part of any other project on the basis of which a course or award was conferred on an earlier occasion on this by any other student(s), also the results of the work are not submitted for the award of any course, degree or diploma within this or in any other University or Institute. We hereby also confirm that all of the experimental work in this report has been done by us.

Belagavi – 590 008
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CERTIFICATE

This is to certify that the project entitled "**BloodBond**" submitted to KLE Technological University's Dr. MSSCET, Belagavi for the partial fulfillment of the requirement for the course – Minor Project-1 (23ECSW303) by Pratham Shinde (02FE22BCS411), Om Kangralkar (02FE22BCS410) and Abhishek Ajatdesai (02FE22BCS402) students in the Department of Computer Science and Engineering, KLE Technological University's Dr. MSSCET, Belagavi, is a bonafide record of the work carried out by them under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any other course completion.

Belagavi – 590 008
Date :

Prof.Savita Bagewadi
(Project Guide)

Prof. Priyanka Patil
(Course Coordinator)

Dr. Rajashri Khanai
(Head of the Department)

Abstract

The abstract presents an overview of potential enhancements for blood bank management software drawn from existing literature. Firstly, the integration of advanced data analytics and machine learning algorithms is highlighted for its ability to improve predictive capabilities in forecasting blood demand and optimizing inventory levels. Secondly, the potential of blockchain technology is explored, offering solutions to enhance transparency, security, and traceability in blood supply chain management. The utilization of Internet of Things (IoT) devices and sensors is discussed for their role in real-time monitoring of blood storage conditions to ensure compliance with regulatory standards and maintain product quality. Additionally, the implementation of mobile applications and telemedicine platforms is examined to facilitate remote donor engagement and accessibility. Moreover, the incorporation of interoperable standards such as Health Level Seven International (HL7) is suggested to streamline data exchange and collaboration among blood banks and healthcare providers. Lastly, improving user interfaces and user experience design is emphasized to enhance usability and efficiency for blood bank staff, donors, and recipients. Overall, these potential enhancements hold promise for optimizing blood bank operations and improving access to safe blood products
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Contents

Chapter 1

Introduction

1.1 Background

Blood bank management plays a critical role in ensuring the availability of safe blood products for medical treatments, emergencies, and surgeries. However, existing blood bank management systems often face challenges in efficiently managing blood inventory, predicting demand, and maintaining transparency in supply chain management.

These challenges can lead to inefficiencies, wastage, and shortages in blood supply, ultimately impacting patient care and public health outcomes. In response, this study explores potential enhancements to existing blood bank management software through a comprehensive literature review.

By leveraging advancements in data analytics, blockchain technology, Internet of Things (IoT) devices, mobile health applications, interoperability standards, and user experience design, blood banks can enhance operational efficiency, ensure regulatory compliance, and improve access to safe blood products. Additionally, these enhancements align with Sustainable Development Goal (SDG) 3: Good Health and Well-being, by contributing to the goal of ensuring access to quality healthcare services and promoting the well-being of all individuals, including through the provision of safe blood transfusions. Through this research, we aim to address key challenges in blood bank management and contribute to the advancement of healthcare systems toward achieving SDG 3 targets

1.2 Problem Statement:

To develop a application for seamless communication between blood donors and hospitals, ensuring timely blood donations and distribution to patients who needs blood in emergency. .

1.3 Objectives:

- Ensure timely access and sufficient blood supplies for medical emergencies and transfusion needs

1.4 SDG Mapping for Objectives

- **Goal 3:** Good Health and Well-being for all.
- **Target 3.8:** Achieve univer- sal health coverage, includ- ing access to safe, effective, quality, and affordable es- sential medicines and vac- cines for al.
- **Indicator 3.8.1:** Universal health coverage implies that al people receive the quality health services they need..

Chapter 2

Literature Survey

- Blood bank management software reveals multiple avenues for improvement. Firstly, the integration of advanced data analytics and machine learning algorithms has shown promise in bolstering predictive capabilities for blood demand forecasting. Researchers have illustrated how machine learning models can effectively predict blood demand by analyzing historical data alongside external factors such as weather and demographic trends. Secondly, the incorporation of blockchain technology has emerged as a potential solution to enhance transparency, security, and traceability in blood supply chain management. Studies advocate for blockchain's use in establishing a decentralized and immutable ledger for tracking blood donations, ensuring the integrity of donor records, and fostering trust among stake holders. Furthermore, the adoption of Internet of Things (IoT) devices and sensors holds promise in providing real-time monitoring of blood storage conditions. This ensures compliance with regulatory standards and safeguards the quality of blood products. Research by Priya et al. (2020) demonstrates the effectiveness of IoT-enabled sensors in monitoring blood storage environments, thereby minimizing spoilage and enhancing overall efficiency. .

- Moreover, the implementation of mobile applications and telemedicine platforms can facilitate remote donor registration, appointment scheduling, and health assessment. This increases donor engagement and accessibility, as highlighted in studies ,ultimately strengthening blood bank operations. Additionally, the adoption of interoperable standards like Health Level Seven International (HL7) can streamline data exchange between different healthcare systems, enhancing collaboration among blood banks and healthcare providers. Research underscores the significance of standardization in improving data sharing and interoperability within the healthcare ecosystem. Lastly, improving user interfaces and user experience design can enhance usability and efficiency for blood bank staff, donors, and recipients. Studies emphasize the importance of adopting user-centered design principles and conducting iterative usability testing to optimize interface design and streamline workflow processes.

- In conclusion, the existing literature offers valuable insights into various modifications and enhancements that can elevate blood bank management software. By harnessing advancements in data analytics, blockchain technology, IoT devices, mobile health applications, interoperability standards, and user experience design, blood banks can bolster operational efficiency, ensure regulatory compliance, and ultimately improve access to safe blood products, potentially saving more lives.

Chapter 3

System Modeling

3.1 Casual Map

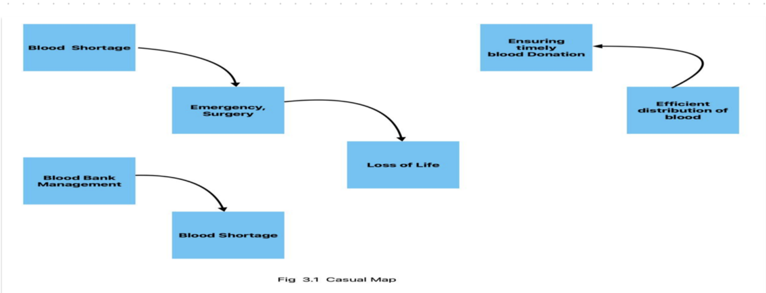


Fig 3.1: Casual Map Diagram

Explanation:

- Blood: Blood, a crucial fluid in the body, carries oxygen and nutrients, and aids in waste removal. It consists of plasma, red and white blood cells, and platelets, with its red color attributed to hemoglobin's oxygen-binding capacity in red blood cells.

- Blood bank management: Blood bank management encompasses the systematic coordination of blood donation, testing, storage, and distribution to ensure a safe and adequate blood supply for medical treatments and emergencies..

- Blood shortage: Blood shortage occurs when demand exceeds supply, risking medical procedures and emergency treatments, often stemming from insufficient blood donations or increased demand.

3.2 Feedback Loop

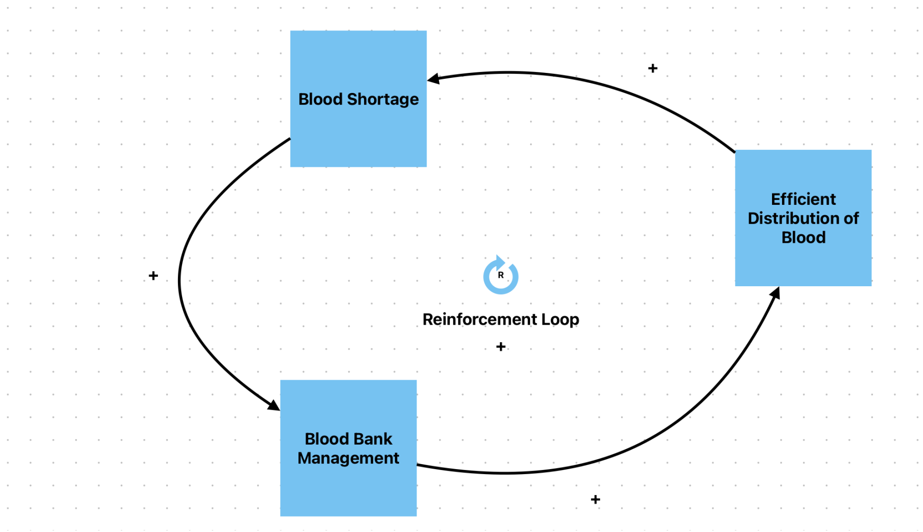


Fig 3.2.a: Reinforcement Loop

Reinforcement Loop:

- Blood shortages: Not enough blood in banks to meet patient needs. Caused by low donations, short shelf life, or high demand for transfusions.
- Efficient blood distribution: Getting blood from donors to hospitals quickly with minimal waste. Relies on good logistics and prioritizes timely delivery..
- Blood bank management: Safe and effective practices for blood collection, testing, storage, and distribution. Blood banks manage inventories, drives, and deliveries to hospitals.

Balancing Loop:

- Blood Storage: Blood banks keep blood products in refrigerators and freezers, but blood has a short shelf life. They need to balance having enough for transfusions without letting it expire.
- Balancing Loop: Blood banks track their supply and schedule drives to avoid shortages. This loop ensures enough blood is available for patients when needed.

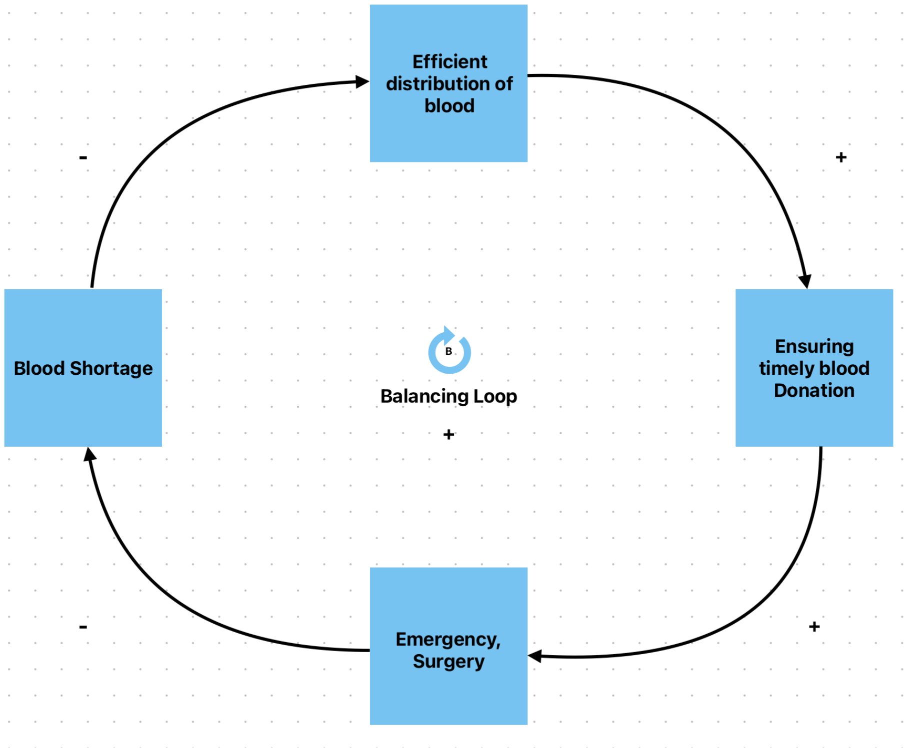


Fig 3.2.b: Balancing Loop

- Engaging Blood Donors: Blood banks rely on volunteers to donate regularly. Public awareness campaigns and donor programs help keep the blood supply safe and adequate.

- Emergency Surgery: Many medical procedures, like surgeries, trauma care, and cancer treatment, rely on blood transfusions available from a healthy blood supply.

3.3 Behavior Over Time [BoT]:

Steps:

- Blood Shortage: The diagram shows a text block labeled "Blood Shortage." This refers to a situation where there isn't enough readily available blood in blood banks to meet patient needs.

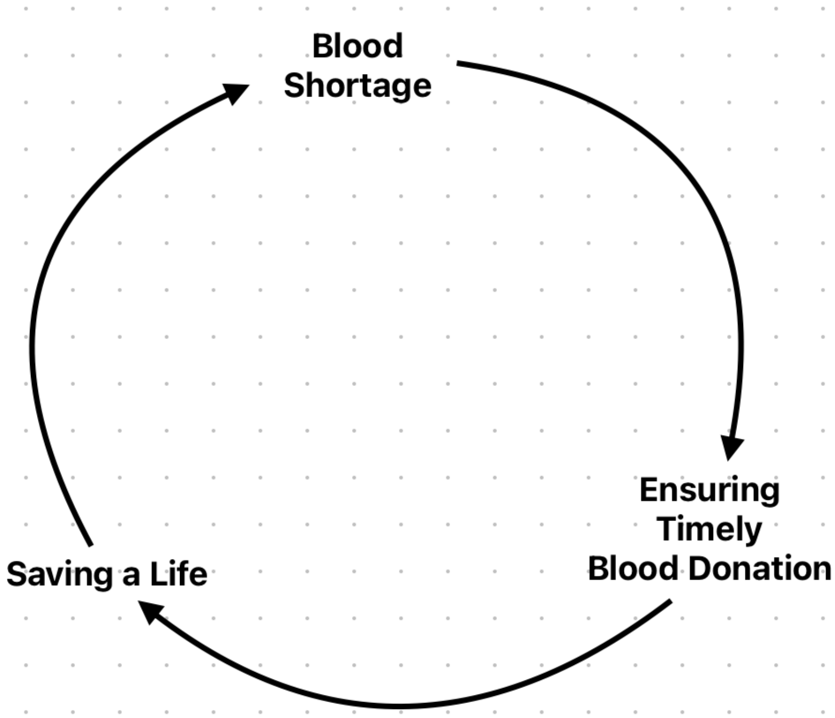
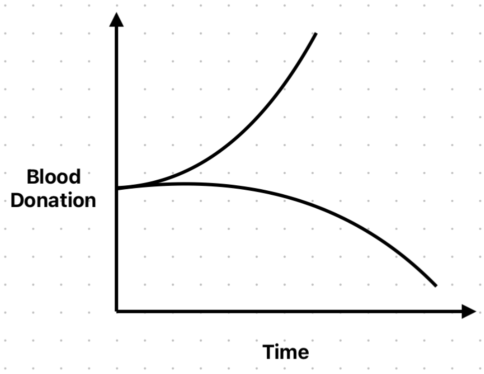
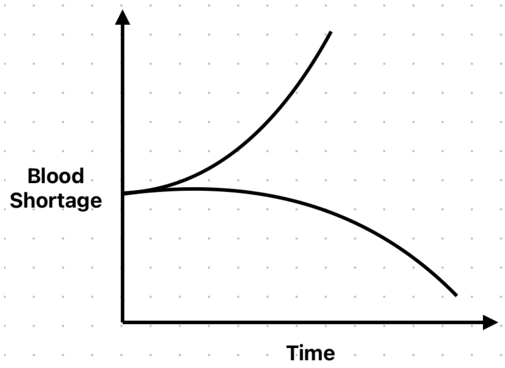


Fig 3.3: Behavior Over Time

- Ensuring Timely Blood Donation: The text block says "Ensuring Timely Blood Donation." This refers to practices that encourage people to donate blood regularly. Blood banks rely on volunteer donors to maintain a safe and sufficient blood supply.
- Saving a Life: The final text block says "Saving a Life." Blood transfusions are a critical part of many medical procedures, and timely blood donations can help save lives.



(A) Water Sanitation



(B) Water Complaints Resolved

Fig 3.4: Behavioral Comparison Over Time

- The graph shows how sanitation levels improve over time. The y-axis might represent the level of sanitation and the x-axis might represent time. The steeper the positive slope on the line, the quicker the sanitation levels improve over time.

- The downward trend in water complaints (blue line) likely indicates improvement in resolving them over time.

Chapter 4

Implementation Details

- **Implementation Platform**
  - **Frontend:** React.js is used for building the user interface of the Blood-Bond. It provides a component-based architecture that enhances the development of interactive and dynamic web applications.
  - **Backend:** Spring Boot is used for creating the backend services. It simplifies the development of ready applications with its embedded server and wide range of libraries.
  - **Database:** MySQL is utilized for managing and storing data. It is a reliable relational database management system that ensures data integrity and supports complex queries.
- **Programming Languages**
  - **JavaScript:** Used for developing the frontend application with React.js.
  - **Java:** Employed for developing the backend services using Spring Boot.
  - **SQL:** Used for querying and managing data in MySQL.
- **Packages and Libraries**
  - **Frontend:**
    - **React.js:** A JavaScript library for building user interfaces.

    - **Axios:** A promise-based HTTP client for making HTTP requests.

  - **Backend:**

    - **Spring Boot:** A framework for building Java-based web applications with built-in support for embedded servers and RESTful services.

    - **Spring Data JPA:** Provides APIs and implementations for data access.

    - **Spring Web:** Provides features to build web applications, including RESTful applications using Spring MVC.

    - **MySQL Connector/J:** A driver that enables Java applications to communicate with MySQL databases.

- **Hardware/Infrastructure Requirements**

  - **Server:** A machine capable of running Spring Boot applications.

  - **Storage:** Adequate disk space for storing application data and user-generated content.

  - **Memory:** Sufficient RAM for handling concurrent users and processing data.

4.1 Testing

DonateBlood Module Test Cases

ID	Description	Expected Result	Actual Result	Result
DTC1	Create new donate blood entry with all fields	The new donate blood entry is saved successfully and returned with an ID and the correct data fields.	The new donate blood entry is saved successfully and returned with an ID and the correct data fields.	Pass
DTC2	Create new donate blood entry with missing field	An error indicating a NOT NULL constraint violation is returned.	An error indicating a NOT NULL constraint violation is not returned.	Fail
DTC3	Read all donate blood entries	All existing donate blood entries are retrieved and displayed correctly.	All existing donate blood entries are retrieved and displayed correctly.	Pass
DTC4	Delete a donate blood entry by ID	The specified donate blood entry is deleted successfully and is no longer present in the list of entries.	The specified donate blood entry is deleted successfully and is no longer present in the list of entries.	Pass

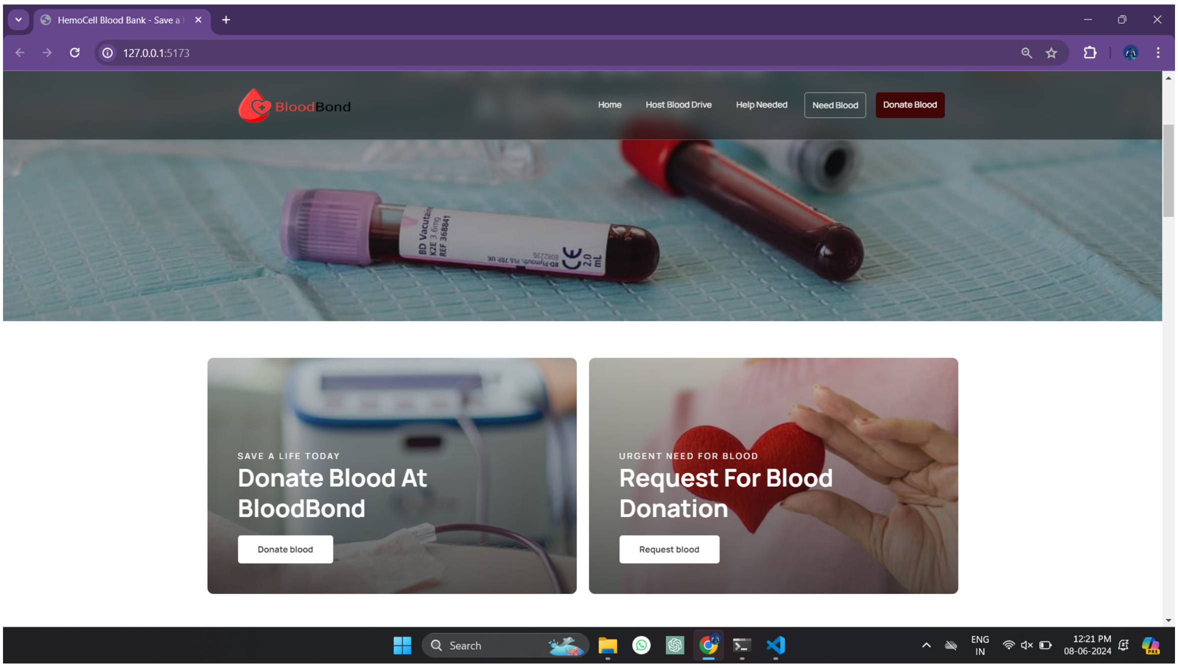
NeedBlood Module Test Cases

ID	Description	Expected Result	Actual Result	Result
NTC1	Create new need blood entry with all fields	The new need blood entry is saved successfully and returned with an ID and the correct data fields.	The new need blood entry is saved successfully and returned with an ID and the correct data fields.	Pass
NTC2	Create new need blood entry with missing field	An error indicating a NOT NULL constraint violation is returned.	An error indicating a NOT NULL constraint violation is not returned.	Fail
NTC3	Read all need blood entries	All existing need blood entries are retrieved and displayed correctly.	All existing need blood entries are retrieved and displayed correctly.	Pass
NTC4	Delete a need blood entry by ID	The specified need blood entry is deleted successfully and is no longer present in the list of entries.	The specified need blood entry is deleted successfully and is no longer present in the list of entries.	Pass

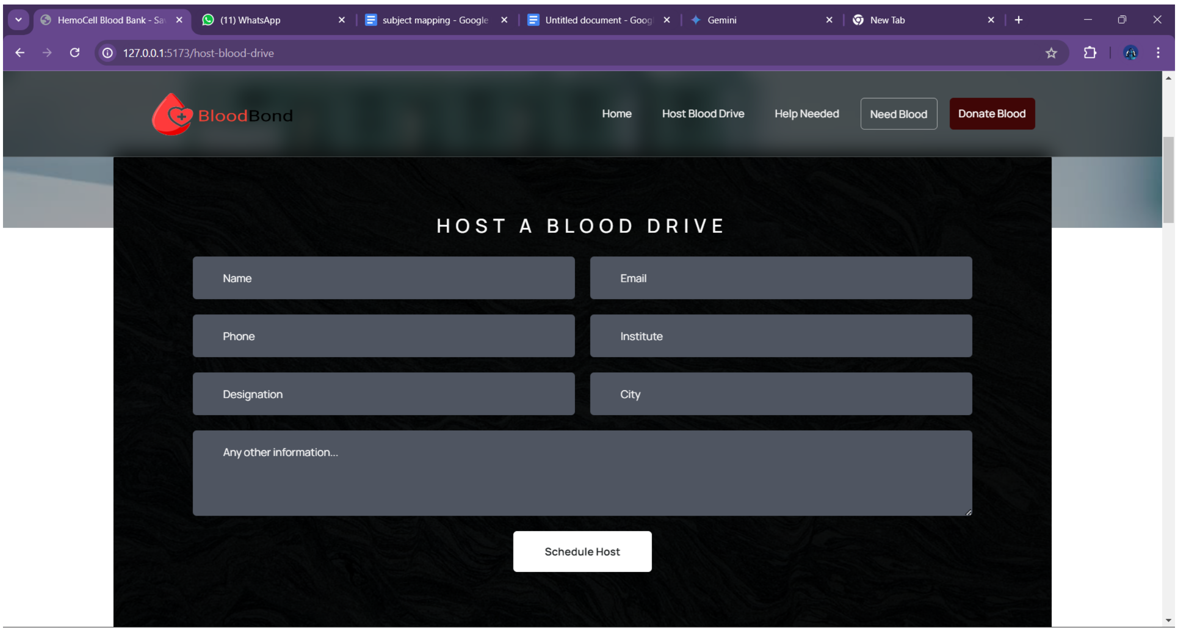
Chapter 5

Results and Outcomes

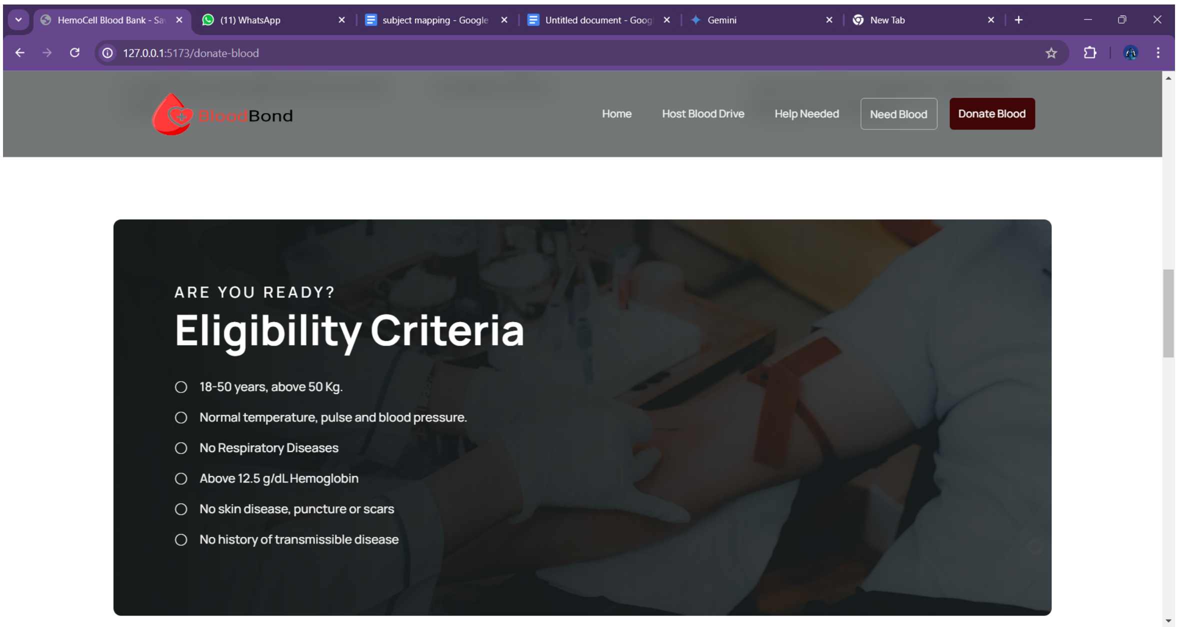
5.1 Interface Design and Database Outcomes



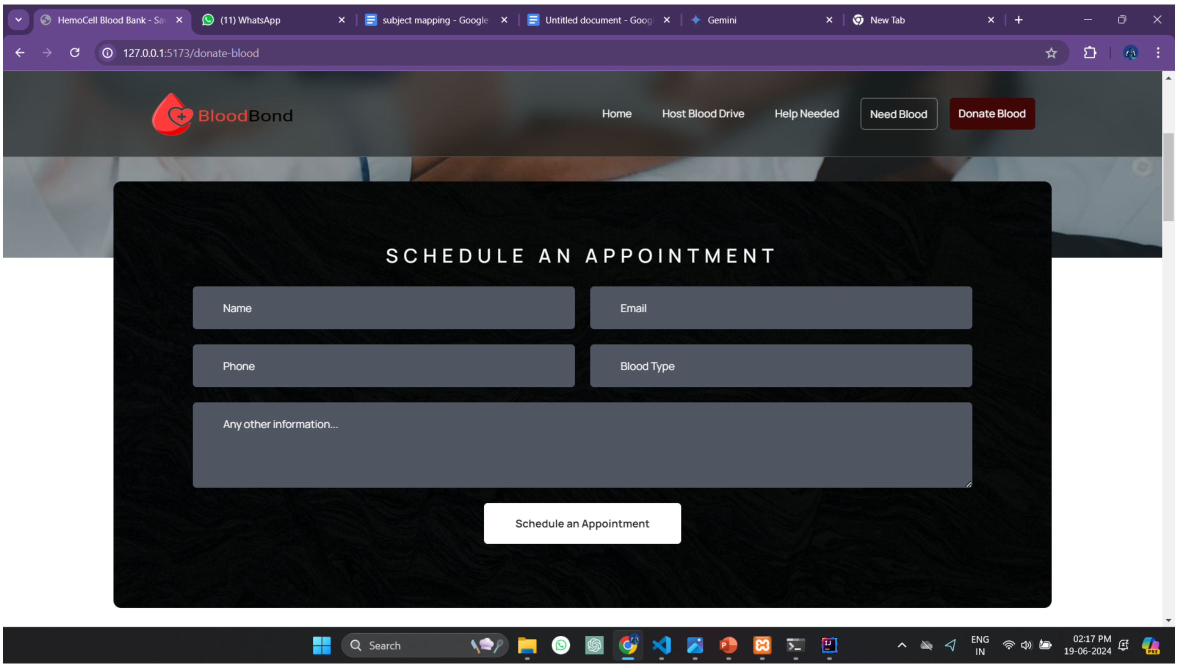
5.1.1 Host blood page



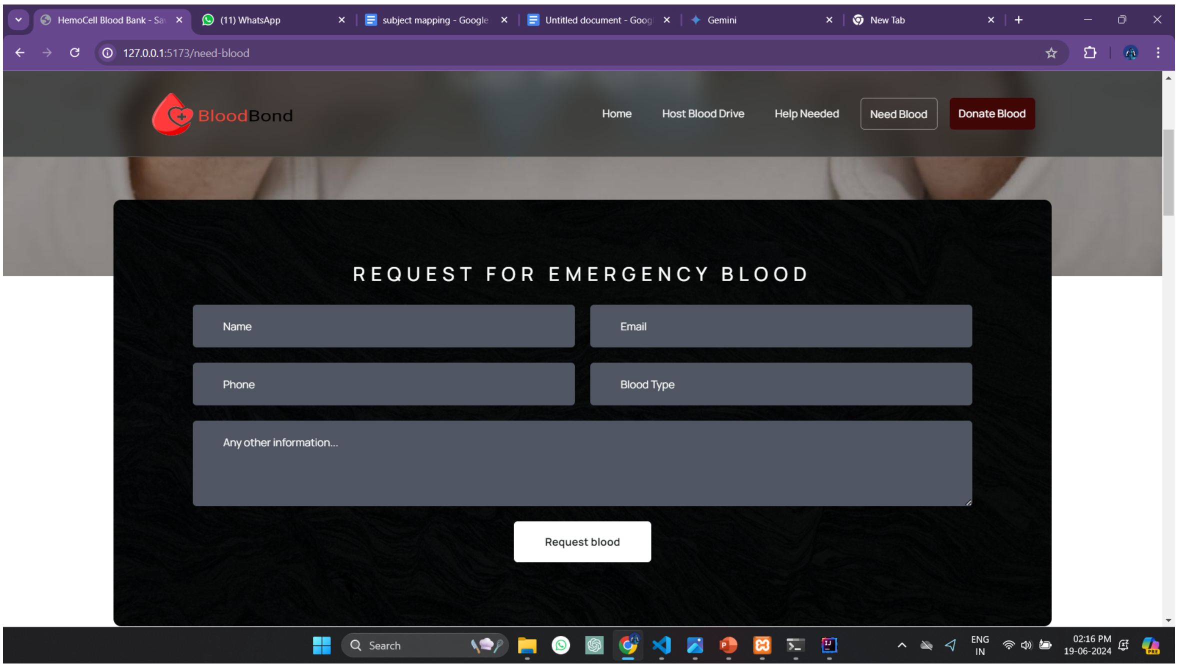
5.1.2 Eligibility Criteria page



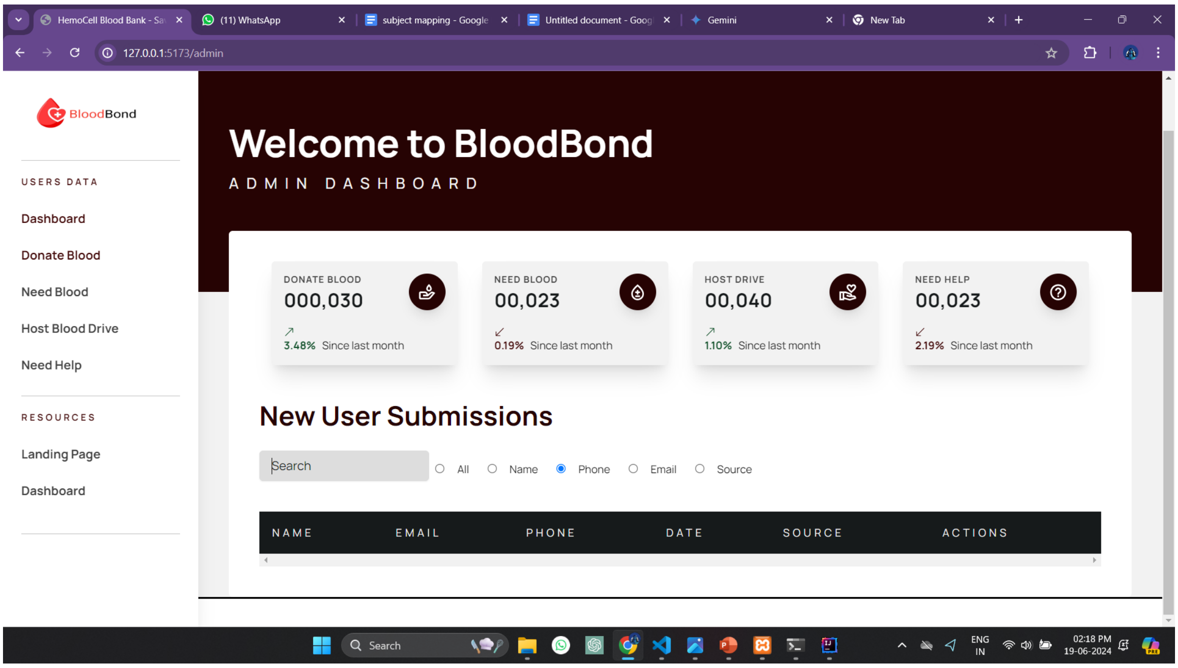
5.1.3 Donate blood page



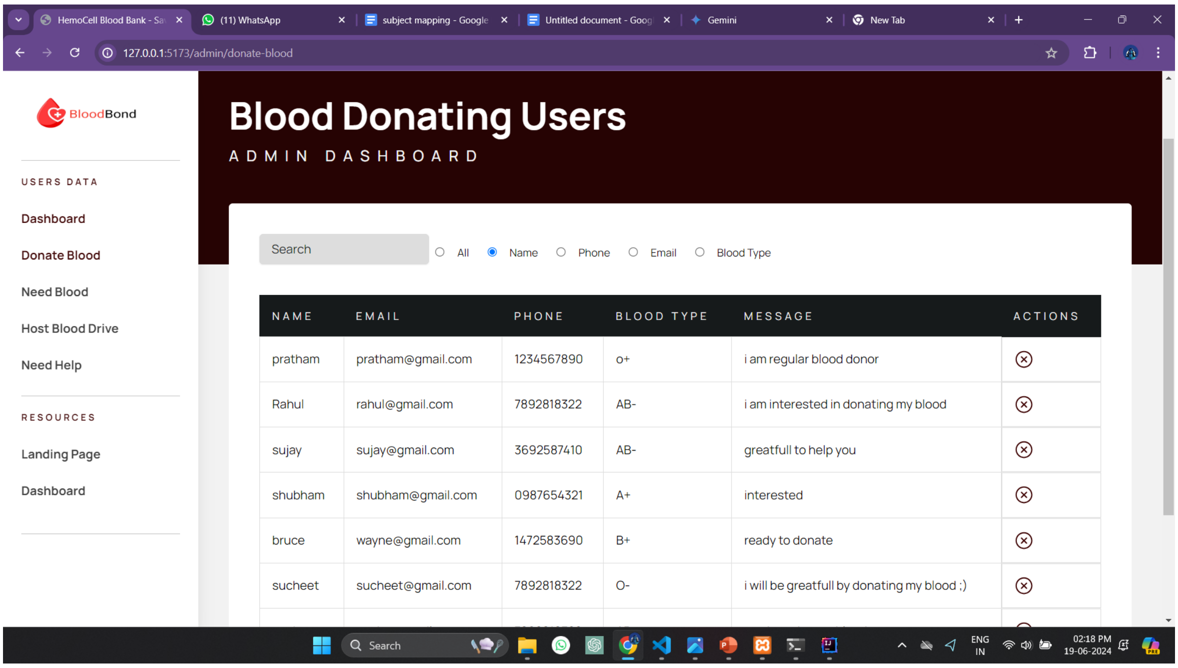
5.1.4 Need blood page



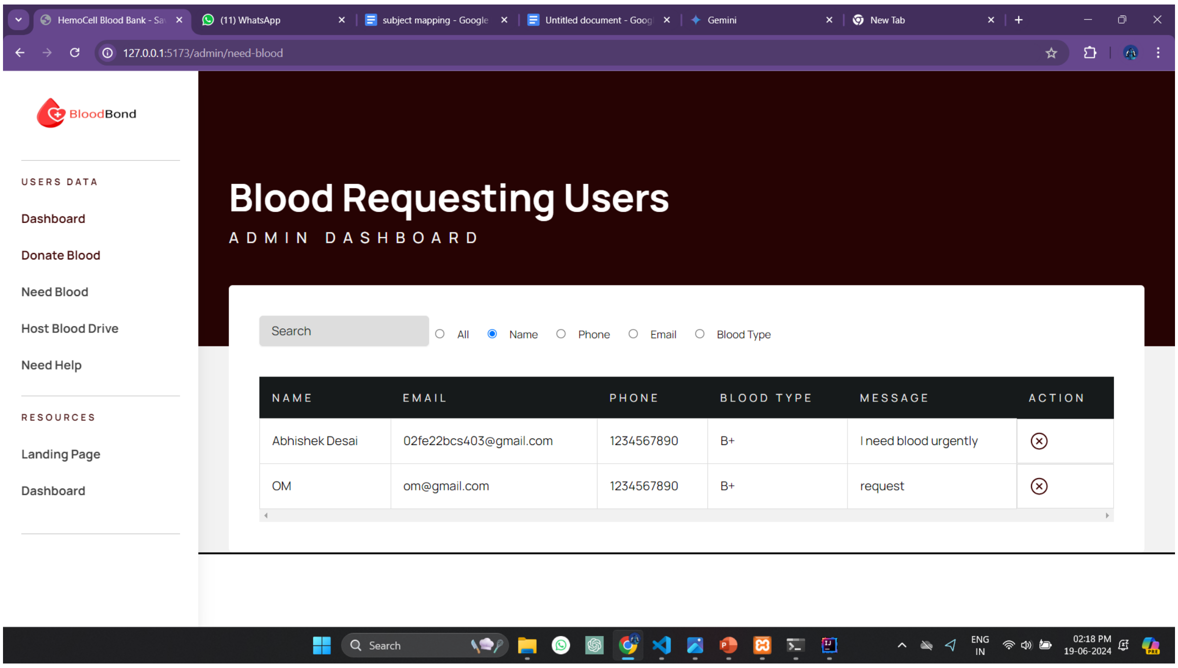
5.1.5 Admin dashboard page



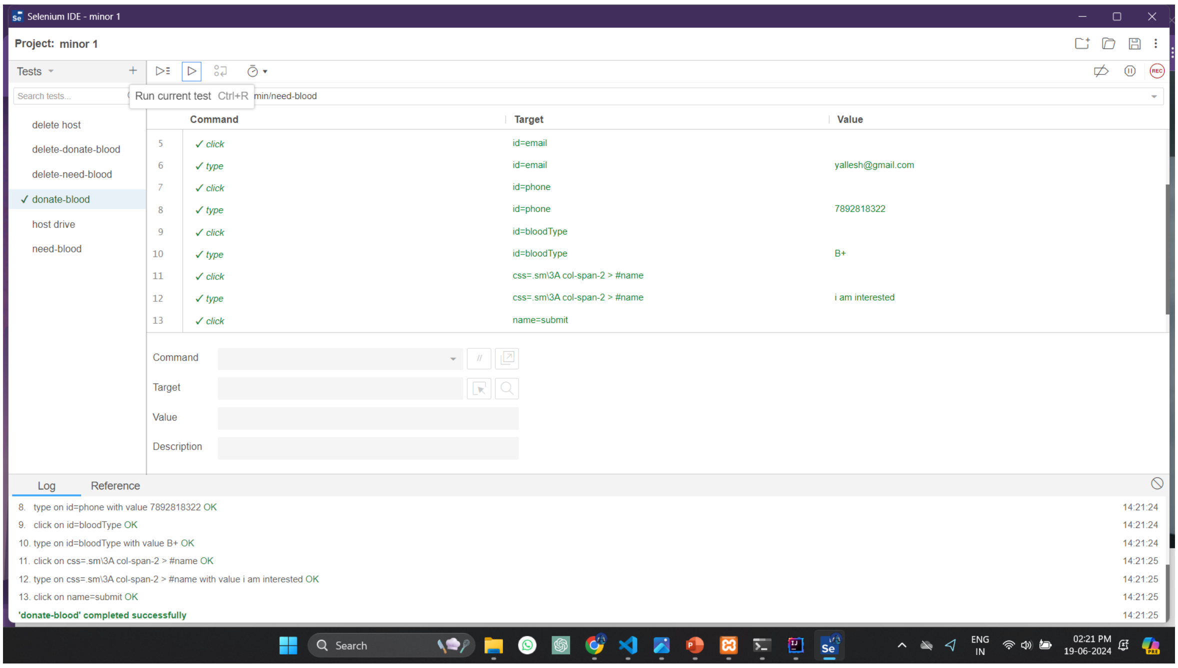
5.1.6 Donate database page



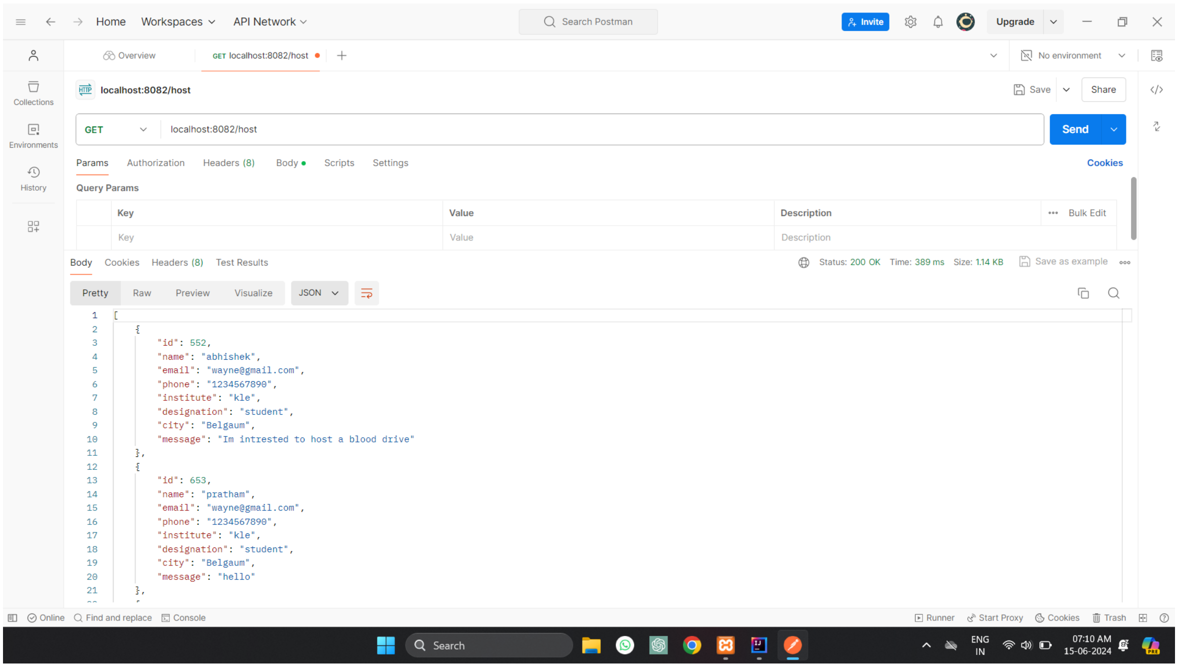
5.1.7 Need database page



5.1.8 Selenium Testing



5.1.9 Backend (Postman)



Conclusions

This project successfully addressed the challenge of promoting good health and well-being for all through a comprehensive BloodBond - Blood Bank Management System platform, aligning with Sustainable Development Goal 3 (SDG 3). By integrating a user- friendly Donor and Receiver page, the platform empowers users to donate blood , host a blood drive and request for blood. The implementation leveraged modern web technologies, including React for the frontend and Spring Boot with MySQL for the backend, providing a robust and scalable architecture. The use of RESTful APIs facilitated seamless communica- tion between the frontend and backend, enhancing the overall user experience.

Stakeholder Visit

(A) KLE Wellness Centre ,4th Gate Udyambaug

(B) KLE Wellness Centre ,4th Gate Udyambaug